

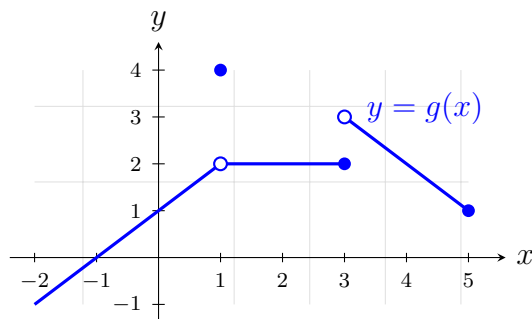
Three Kinds of Evidence About a Limit

Reading limits from a graph, estimating them from a table of values, and confirming them with algebra

Before you start. A limit claim can be supported three ways, and a good answer says which one you used.

- **Graphical** — follow the curve in from each side and see what height it heads for. A filled dot tells you the function *value*; it does not tell you the limit.
- **Numerical** — evaluate at inputs closing in from both sides. A table suggests a limit; it never proves one.
- **Symbolic** — factor, rationalize, or clear the compound fraction, then substitute. This is the one that settles the matter.

Graph of g — used by problems 3 and 6 only. Open circles are excluded points; filled dots are included.



1. Evaluate $\lim_{x \rightarrow 3} (x^2 - 2x + 5)$, and state in one phrase why direct substitution is allowed here.

Answer: _____

x	2.9	2.999	3.001	3.1
$f(x)$				

Answer: _____

Answer: _____

4. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$. Substitution gives $\frac{0}{0}$, so it tells you nothing — factor first.

Answer: _____

5. Let $f(x) = \frac{x^3 - 8}{x - 2}$. Complete the table, state the limit it suggests, then confirm that limit algebraically.

x	1.9	1.99	2.01	2.1
$f(x)$				

Answer: _____

6. Use the graph of g above. Give $\lim_{x \rightarrow 3^-} g(x)$ and $\lim_{x \rightarrow 3^+} g(x)$, and explain why $\lim_{x \rightarrow 3} g(x)$ does not exist. Given that $g(x) = 6 - x$ on the branch to the right of 3, confirm the right-hand limit algebraically.

Answer: _____

7. Evaluate $\lim_{x \rightarrow 0} \frac{\frac{1}{x+2} - \frac{1}{2}}{x}$. Combine the two fractions in the numerator before you do anything else.

Answer: _____

8. The function $f(x) = \frac{x^2 + ax + 3}{x - 1}$ has a finite limit as $x \rightarrow 1$.

(a) The numerator must be zero at $x = 1$ for that to happen. Use this to find a .

(b) With that value of a , evaluate $\lim_{x \rightarrow 1} f(x)$.

Answer: _____

9. Let $f(x) = \frac{|x - 4|}{x - 4}$. Complete the table, then explain why $\lim_{x \rightarrow 4} f(x)$ does not exist even though both one-sided limits do.

x	3.9	3.99	4.01	4.1
$f(x)$				

Answer: _____

10. Synthesis. Let $p(x) = \frac{x^2 + x - 6}{x - 2}$ for $x \neq 2$, with $p(2)$ not yet defined.

(a) Evaluate p at $x = 1.9$ and at $x = 2.1$.

(b) Find $\lim_{x \rightarrow 2} p(x)$ algebraically, and state the value of $p(2)$ that would make p continuous at 2.

(c) Sketch the graph of p near $x = 2$, marking the hole, and write one or two sentences saying how the table, the algebra and your sketch agree.

Answer: _____